

Problem 1

1. This MAC is secure since it uses the PRF F . The outer PRF returns a deterministic, pseudorandom string and is therefore also a secure MAC. This means that the attacker's probability of generating a forgery is $\frac{1}{2^n}$, resulting in a negligible advantage.
2. This scheme is secure. Since m is appended with different bit strings ($1^{n/2}$ and $0^{n/2}$ for the first and second halves respectively), the secure pseudorandom strings generated by each F_k cannot be interchanged. Namely, $m||1^{n/2}$ and $m||0^{n/2}$ will always be different. Even if an attacker tries to rearrange computed tags from the oracle, since the appended bits are strictly placed, there is no way the attacker can create a forgery. Thus, the probability of generating one is $\frac{1}{2^n}$, resulting in a negligible advantage.
3. This scheme is insecure. For an attack, an attacker can query the MAC oracle for a tag over a message of $n - 2$ bits of 1s and append 01. This results in a tag t , with $F_k(\overline{m})$ being computed. This is also a correct tag over a message of $n - 2$ bits of 0s, appended with 10 (resulting in $F_k(m)$ being evaluated, resulting in a tag that equals t). Therefore, the probability of making a correct forgery is 1, and the advantage of the attacker is $1 - \frac{1}{2^n}$, which is non-negligible. Therefore, the probability of making a correct forgery is 1, and the advantage of the attacker is $1 - \frac{1}{2^n}$, which is non-negligible.

Problem 2

1. h' is not CRHF. Since h_2 is not CRHF (even though it is OWF), an attacker can find a collision via messages m and m' , where $m = m_1||m_2$ and $m' = m_1||m_3$ such that $h_2(m_2)$ and $h_2(m_3)$ collide (since it is not guaranteed that it is hard to find a collision for a OWF, and m_2 and m_3 are not random, being chosen by the attacker). Since an attacker can find a collision for h' , the success probability is 1 and the attacker's advantage is $1 - \frac{1}{2^n}$, which is non-negligible.
2. h'' is OWF, since the outermost layer is h_2 , which is also OWF. In the OWF game, an attacker does not get the outputs for h_1 (which constitutes the input for h_2) and therefore cannot find a collision, resulting in a success probability of $\frac{1}{2^n}$, which is negligible.
However, h'' is not CRHF. Since the outermost layer is OWF but not CRHF, it is not hard for an attacker to find a collision. Since they are playing the CRHF, they can just exhaustively search messages until they find a collision for h_2 . Therefore, the success probability is $1 - \frac{1}{2^n}$, which is non-negligible.
3. Sponge's proposal will preserve the security. Since h_1 is already CRHF and is the outermost layer, it will preserve being CRHF which is required for the application. Alice's proposal, using h_2 as the outermost layer, is not CRHF. An attacker, in the CRHF game, can simply brute force inputs to h_2 and find a collision since it is not CRHF. This results in a success probability of 1, with a non-negligible advantage of $1 - \frac{1}{2^n}$.
4. (a) The number of nodes at the leaf level is 32 (one node for each file and its hash). The total size is 47 (32 leaf nodes, 5 layers of hashes including the leaf nodes and the root node).
(b) The proof of inclusion is $h_{1-8}, h_{9-12}, h_{14}$, and h_{15-16} . The size of this is 5. Eve can verify this PoI by computing $h(h(f_{13})||h_{14})$ to get h_{13-14} , then hash that with h_{15-16} to get h_{13-16} . Then, hash that with h_{9-12} to get h_{9-16} , and then finally checking if that hashed with h_{1-8} is the same as the root node of the Merkle Tree.